

ePBL Internship Report

CampusHub: A Full Stack Web-Based College Management System

Domain: Full-Stack Web Development

Submitted By

Roll No.	Name
23881A04H3	Mustigolla Rishith Yadav
23881A04G4	Konda Shashi Kumar
23881A04D5	Banoth Akhil Nayak

**Department of Electronics and Communication
Engineering**

Vardhaman College of Engineering

Guide

Dr.Purna Chandar Reddy
Assistant Professor

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1. Abstract

CampusHub is a web-based College Management System developed using PHP, MySQL, HTML, CSS, Bootstrap, and JavaScript. The application provides separate portals for administrators, faculty members, and students. It simplifies academic management by providing secure authentication, student and faculty management, attendance tracking, CGPA management, and a centralized notice board. The project demonstrates practical implementation of Full Stack Web Development concepts along with database integration and session management.

2. Introduction

Educational institutions manage large volumes of academic data such as student records, faculty information, attendance, and notices. Traditional manual systems are time-consuming and prone to errors.

CampusHub was developed to provide a centralized digital platform for managing academic information efficiently. The system offers role-based access, allowing administrators, faculty, and students to perform their respective tasks securely through a web interface. The project follows a modular architecture and integrates frontend technologies with PHP and MySQL to create a complete Full Stack Web Application.

3. Objectives

- Develop a secure College Management System.
- Implement role-based login for Admin, Faculty, and Students.
- Manage student and faculty records.
- Maintain student attendance and CGPA.
- Display notices through a centralized notice board.
- Demonstrate Full Stack Web Development concepts.

4. Technology Stack

Technology	Purpose
HTML5	Web Structure
CSS3	Styling
Bootstrap 5	Responsive Design
JavaScript	Client-side Interaction
PHP	Backend Development
MySQL	Database Management
XAMPP	Local Server
Git & GitHub	Version Control

5. System Architecture

CampusHub follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer.

- **Presentation Layer:** Developed using HTML, CSS, Bootstrap, and JavaScript to provide an interactive and responsive user interface.
- **Application Layer:** Implemented using PHP, which handles user authentication, business logic, CRUD operations, session management, and communication with the database.
- **Database Layer:** MySQL is used to store and manage student, faculty, administrator, and notice information securely.

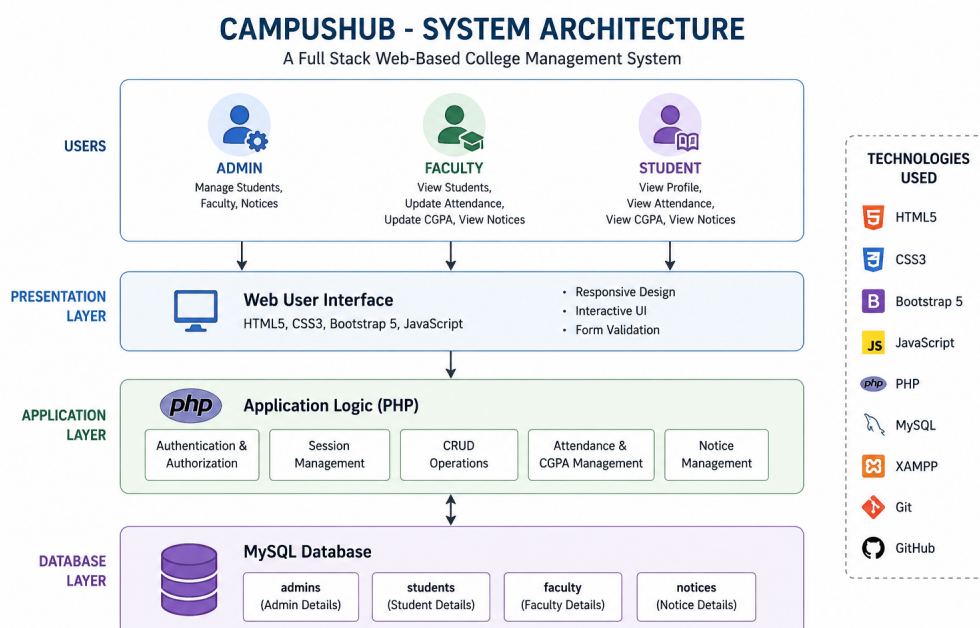


Figure 1: System Architecture of CampusHub

6. Database Design

CampusHub uses MySQL as the backend database for storing application data. The database consists of the following major tables.

Table	Purpose
Admins	Stores administrator login credentials.
Students	Stores student personal and academic information.
Faculty	Stores faculty details and login credentials.
Notices	Stores notices displayed on all dashboards.

The database design ensures data consistency and enables efficient retrieval and updating of records.

7. Modules

7.1 Admin Module

The Admin Module provides complete control over the application.

Major functionalities include:

- Secure Admin Login
- Dashboard with Statistics
- Add, Edit, Delete, and Search Students
- Add, Edit, Delete, and Search Faculty
- Add and Delete Notices
- Logout

7.2 Faculty Module

Faculty members can manage academic information assigned to them.

Major functionalities include:

- Secure Faculty Login
- Faculty Dashboard
- View Student Details
- Update Student CGPA
- Update Student Attendance Percentage
- View Notices
- Logout

7.3 Student Module

Students can access their academic information through a dedicated dashboard.

Major functionalities include:

- Secure Student Login
- Student Dashboard
- View Personal Profile
- View CGPA
- View Attendance Percentage
- View Notices
- Logout

8. Key Features

The major features implemented in CampusHub are:

- Role-Based Authentication
- Secure Session Management
- Student Management
- Faculty Management
- Attendance Management
- CGPA Management
- Notice Board
- CRUD Operations
- Responsive Bootstrap Interface
- MySQL Database Connectivity
- Git and GitHub Version Control

9. Implementation

CampusHub was developed using the Full Stack Web Development approach by integrating frontend technologies with backend programming and a relational database.

Frontend Development

The user interface was designed using HTML5, CSS3, Bootstrap 5, and JavaScript. Bootstrap was used to create responsive webpages that can adapt to different screen sizes. JavaScript was used to improve user interaction and validation.

Backend Development

PHP was used as the server-side scripting language to implement the application's business logic. PHP handles user authentication, session management, CRUD operations, and communication with the MySQL database.

Database Integration

MySQL was used to store all application data. PHP establishes a secure connection with the database using MySQLi, allowing users to add, update, delete, and retrieve records efficiently.

10. Project Screenshots

The following screenshots demonstrate the successful implementation of the CampusHub application.

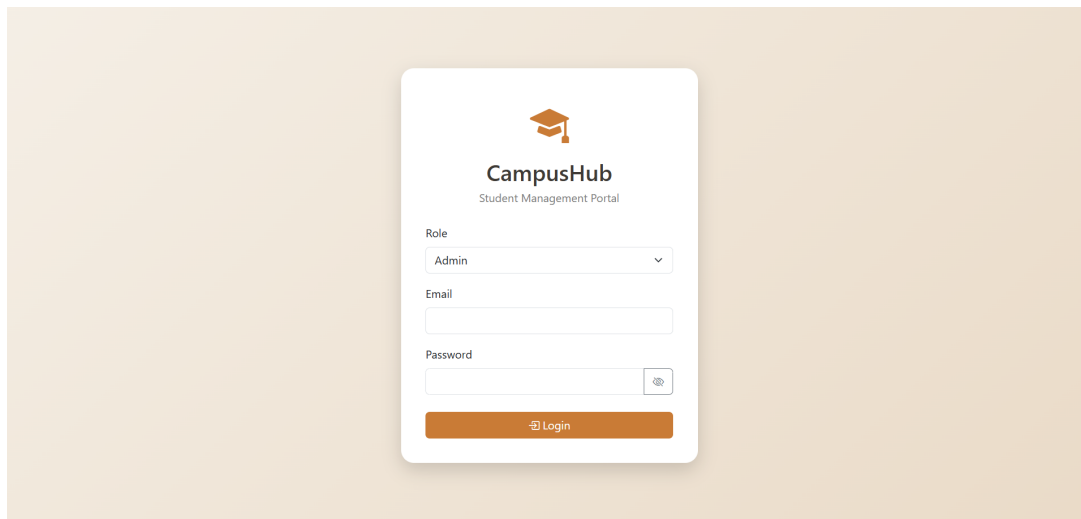


Figure 2: CampusHub Login Page

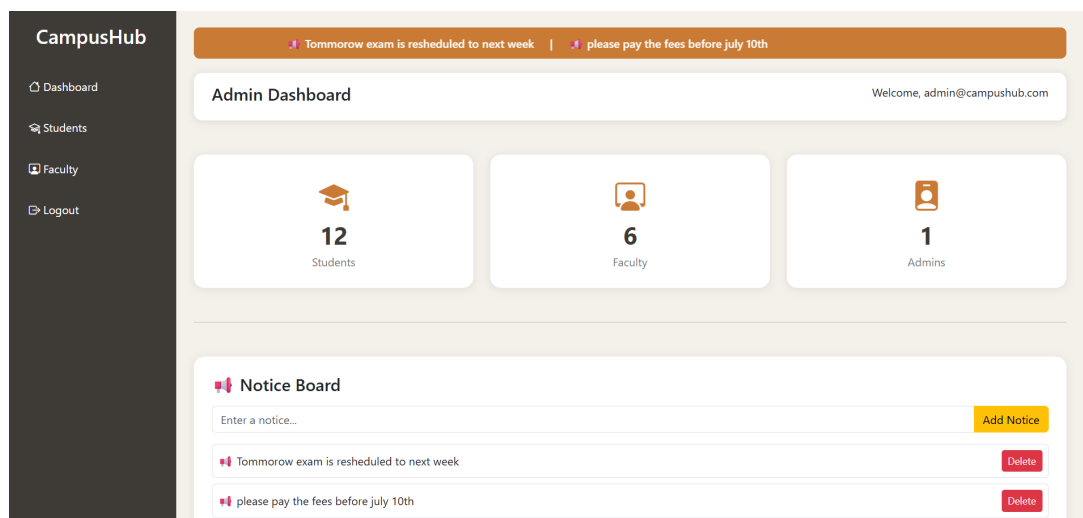


Figure 3: Admin Dashboard

 Student Management Add Student							
Search by Roll No or Name Search							
ID	Roll No	Name	Email	Department	Year	CGPA	Action
3	55	bunny	bunny@gmail.com	CSE	3	8.86	 
6	87	Rakesh	rakesh@gmail.com	IT	4	7.85	 
7	34	Sunny	sunny@gmail.com	CSD	3	9.10	 
8	75	Sai	sai@gmail.com	ECE	4	6.78	 
9	2	Suresh	suresh@gmail.com	ECE	2	8.12	 
10	15	Ramesh	ramesh@gmail.com	EEE	3	6.54	 
11	69	Venkat	venkat@gmail.com	CIV	4	9.75	 
12	12	Pavan	pavan@gmail.com	MEC	3	8.90	 
13	90	Anisha	anisha@gmail.com	ECE	2	8.88	 
14	17	Priya	priya@gmail.com	CSE	4	7.40	 
15	23	Lakshmi	lakshmi@gmail.com	IT	3	7.75	 

Figure 4: Student Management

 Faculty Management Add Faculty						
Search by Faculty ID, Name, Email, Department or Designation Search						
ID	Faculty ID	Name	Email	Department	Designation	Action
1	A1	Dr.Virendar reddy	virendar@gmail.com	ECE	Associate professor	 
3	A5	P.Shiva	shiva@gmail.com	CIV	Assistant professor	 
4	A2	Dr.Priyanka	priyanka@gmail.com	CSE	Associate professor	 
5	A3	M.Raju	raju@gmail.com	IT	Assistant professor	 
6	A4	D.Ganesh	ganesh@gmail.com	CSD	Associate professor	 
7	A6	Dr.Bhupathi	bhupathi@gmail.com	MEC	Associate professor	 

Back to Dashboard

Figure 5: Faculty Management

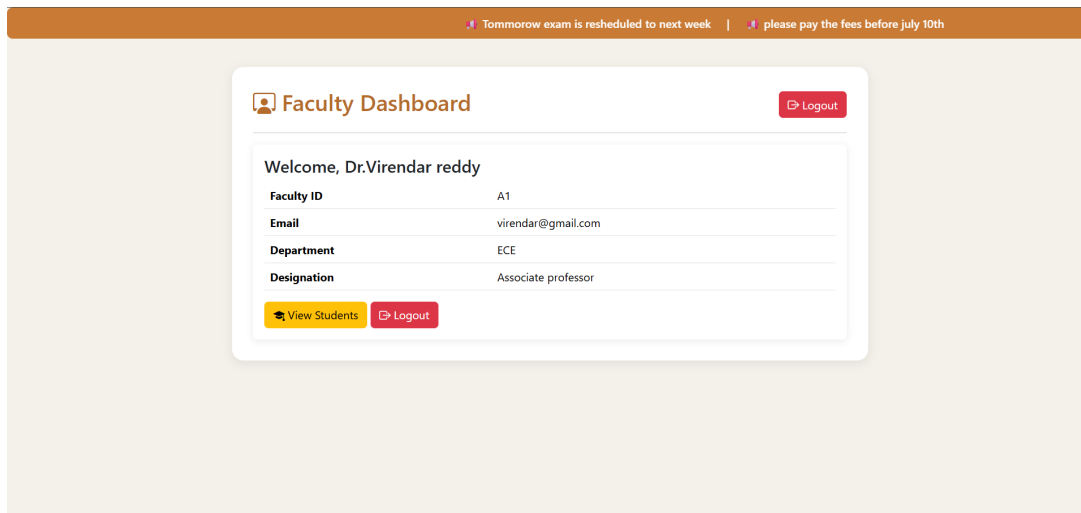


Figure 6: Faculty Dashboard

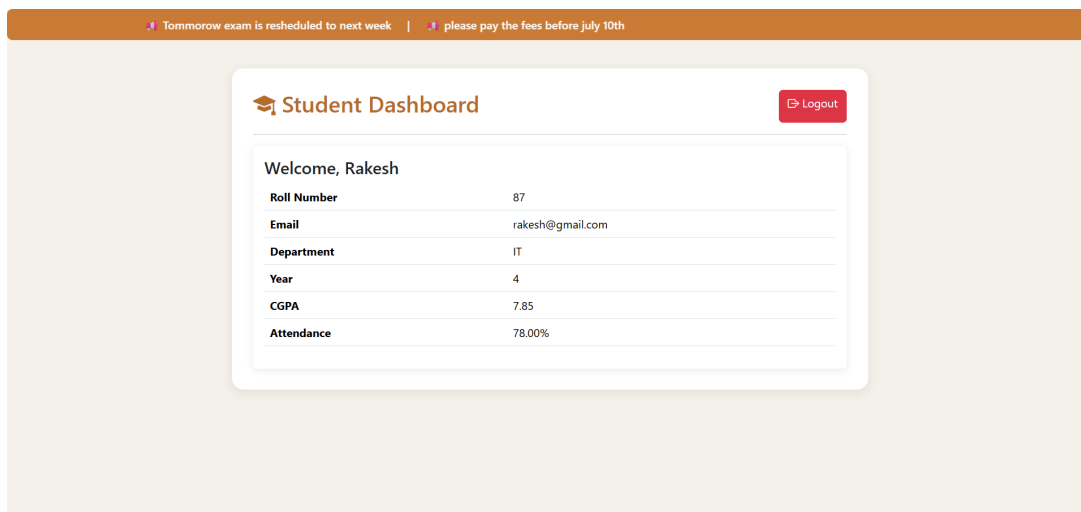


Figure 7: Student Dashboard

11. Results

The CampusHub application was successfully developed and tested in a local environment using XAMPP and MySQL. All modules performed as expected.

The project achieved the following outcomes:

- Secure login for Admin, Faculty, and Students.
- Student record management using CRUD operations.
- Faculty record management.
- Attendance percentage management.
- CGPA management.

- Centralized notice board.
- Responsive dashboards.
- Database connectivity using MySQL.
- Version control using Git and GitHub.

12. Future Enhancements

Although CampusHub fulfills the current requirements, several additional features can be incorporated in future versions.

- Password encryption using `password.hash()`.
- Email notification system.
- Student profile image upload.
- PDF report generation.
- Advanced search and filtering.
- Analytics dashboard with charts.
- Cloud deployment.
- Mobile application support.

13. Conclusion

CampusHub successfully demonstrates the practical implementation of Full Stack Web Development technologies in building a College Management System. The project integrates PHP, MySQL, HTML, CSS, Bootstrap, and JavaScript to provide a secure and user-friendly platform for managing academic information.

The developed system simplifies student and faculty management while improving communication through a centralized notice board. The project also demonstrates concepts such as authentication, session management, CRUD operations, responsive web design, and database integration.

Overall, CampusHub serves as an efficient academic management solution and provides valuable practical experience in Full Stack Web Development.

14. References

References

- [1] PHP Documentation. <https://www.php.net>
- [2] MySQL Documentation. <https://dev.mysql.com/doc/>
- [3] Bootstrap Documentation. <https://getbootstrap.com>
- [4] JavaScript Documentation. <https://developer.mozilla.org>
- [5] GitHub Documentation. <https://docs.github.com>